role_id
role_id
user_id
user_id
user_id
backup_id
requested_by
author_id
customer_id
customer_id
customer_id
created_by
created_by
product_id
product_id
quotation_id
quotation_id
id

ROLE

integer	id	PK	
role_type	type	UK	SUPERADMIN, ADMIN, OPERATOR

BACKUP_FILE

uuid	id	PK	
varchar	file_name		
timestamp	created_at		
double	size_mb		
backup_type	type		DAILY, WEEKLY, MONTHLY
varchar	sha256_hash		
integrity_status	integrity_status		VALID, CORRUPTED, NOT_VERIFIED
varchar	storage_path		
boolean	encrypted		

ROLE_PERMISSION

integer	role_id	PK,FK	
permission	permission	PK	USER_MANAGE, BACKUP_MANAGE, BACKUP_RESTORE, CATALOG_VIEW, CATALOG_EDIT, CUSTOMER_MANAGE, QUOTATION_MANAGE, EXPORT_QUEUE_MANAGE, WORK_ORDER_MANAGE

APP_USER

uuid	id	PK	
integer	role_id	FK	
varchar	full_name		
varchar	email	UK	Login key, read-only after creation
varchar	job_title		
varchar	password_hash		bcrypt
account_status	status		ACTIVE, INACTIVE
timestamp	last_access_at		

CUSTOMER

uuid	id	PK	
customer_type	type		NATURAL_PERSON, LEGAL_ENTITY
varchar	tax_id	UK	ID number or NIT, read-only
varchar	name		fullName or businessName
varchar	address		
varchar	primary_phone		
varchar	email		
varchar	commercial_contact		
customer_status	status		ACTIVE, INACTIVE

TOTP_ENROLLMENT

uuid	id	PK	
uuid	user_id	FK,UK	Max one enrollment per user (0..1)
varchar	secret		
bytea	qr_code		
timestamp	issued_at		
timestamp	expires_at		issued_at + 5 min
enrollment_status	status		PENDING, CONFIRMED, EXPIRED

SESSION

uuid	id	PK	
uuid	user_id	FK	
varchar	access_token_id		
varchar	refresh_token_id		
timestamp	created_at		
timestamp	last_activity_at		Inactivity timeout 15 min
session_status	status		ACTIVE, TERMINATED
termination_reason	termination_reason		LOGOUT, INACTIVITY, NEW_LOGIN, ACCOUNT_DEACTIVATED

AUDIT_ENTRY

uuid	id	PK	Immutable
uuid	user_id	FK	Required
timestamp	timestamp		
audit_action	action		CREATED, UPDATED, STATUS_CHANGED, APPROVED, REJECTED, CANCELLED, EXPORTED
varchar	entity_type		
varchar	entity_id		
text	details		

RESTORE_PROCESS

uuid	id	PK	
uuid	backup_id	FK	
uuid	requested_by	FK	
restore_status	status		VERIFYING, VERIFIED, VERIFICATION_FAILED, CONFIRMED, COMPLETED
boolean	verification_success		VerificationResult
varchar	verification_message		VerificationResult
timestamp	verified_at		VerificationResult
timestamp	started_at		
timestamp	finished_at		

COMMERCIAL_NOTE

uuid	id	PK
uuid	customer_id	FK
uuid	author_id	FK
text	content	
timestamp	created_at	
timestamp	updated_at	

PRODUCT

uuid	id	PK	
varchar	code	UK	Read-only
varchar	name		
varchar	unit_of_measure		
product_status	status		ACTIVE, INACTIVE

WORK_ORDER

uuid	id	PK	
bigint	order_number	UK	From database sequence, read-only
uuid	customer_id	FK	
uuid	created_by	FK	
date	agreed_delivery_date		
work_order_status	status		PENDING, REGISTERED_IN_SIIGO, RECONCILED, CANCELLED
text	cancellation_reason		Required when cancelling
timestamp	created_at		

QUOTATION

uuid	id	PK	
uuid	customer_id	FK	Nullable: chatbot requester may not be a customer yet
timestamp	issued_at		
varchar	requester_full_name		RequesterContact
varchar	requester_email		RequesterContact
varchar	requester_phone		RequesterContact
decimal	estimated_total_min		PriceRange.min
decimal	estimated_total_max		PriceRange.max
varchar	currency		
varchar	commercial_terms		
date	valid_until		
date	estimated_dispatch_date		
dispatch_date_source	dispatch_date_source		SUGGESTED, MANUAL
quotation_status	status		PENDING_REVIEW, APPROVED, MODIFIED, REJECTED
text	rejection_reason		Required when rejecting

EXPORT_BATCH

uuid	id	PK
uuid	created_by	FK
varchar	file_name	
timestamp	created_at	

QUOTATION_ITEM

uuid	id	PK	
uuid	quotation_id	FK	
uuid	product_id	FK	
varchar	product_code		Snapshot
varchar	description		Snapshot
integer	quantity		
decimal	unit_price_min		PriceRange.min (estimated)
decimal	unit_price_max		PriceRange.max (estimated)

WORK_ORDER_ITEM

uuid	id	PK
uuid	work_order_id	FK
uuid	product_id	FK
integer	quantity	

PROCESSING_QUEUE_ENTRY

uuid	id	PK	
uuid	quotation_id	FK,UK	Approved quotation
uuid	export_batch_id	FK	Null until exported
export_status	export_status		PENDING_EXPORT, EXPORTED
timestamp	enqueued_at		